Lesson 13: Transformations of Graphs — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Describe the transformation $y = f(x) + 6$.

2. Describe $y = f(x) - 3$.

3. Describe $y = f(x - 5)$.

4. Describe $y = f(x + 2)$.

5. Describe $y = -f(x)$.

6. Describe $y = f(-x)$.

Section 2 — Core practice

7. Describe $y = 3f(x)$.

8. Describe $y = f(2x)$.

9. Describe the two transformations in $y = f(x - 4) + 1$.

10. Describe the two transformations in $y = 2f(x) - 5$.

11. Describe $y = \frac{1}{2} f(x)$.

12. Describe $y = f(x + 3) - 2$.

Section 3 — Challenge & exam-style

13. The point $(2, 5)$ is on $y = f(x)$. Find its image on $y = f(x - 1) + 4$.

14. The point $(-1, 3)$ is on $y = f(x)$. Find its image on $y = 2f(x)$.

15. The point $(4, 8)$ is on $y = f(x)$. Find its image on $y = f(2x)$.

16. Describe fully the transformations mapping $y=f(x)$ to $y = -f(x) + 3$.